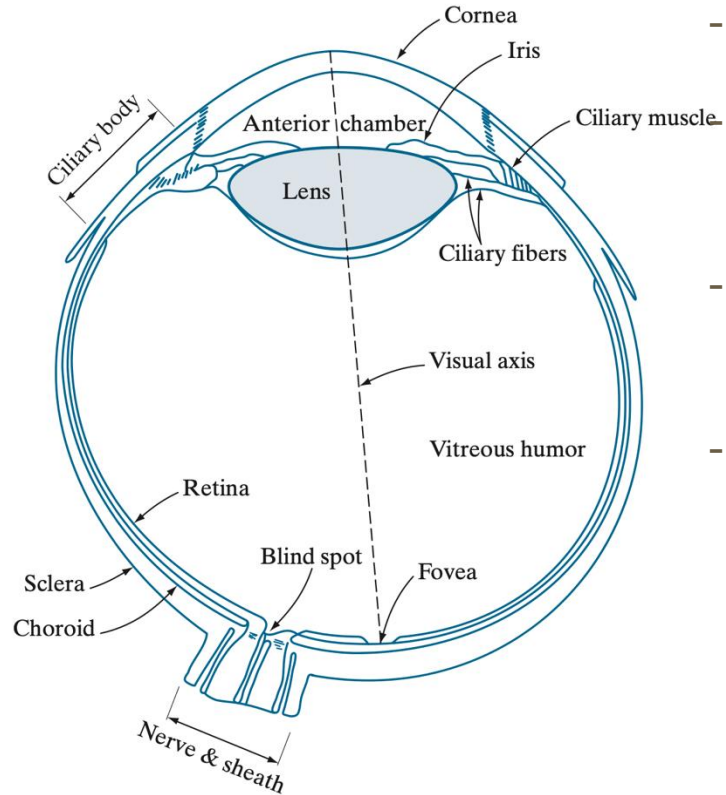

Digital Image Fundamentals

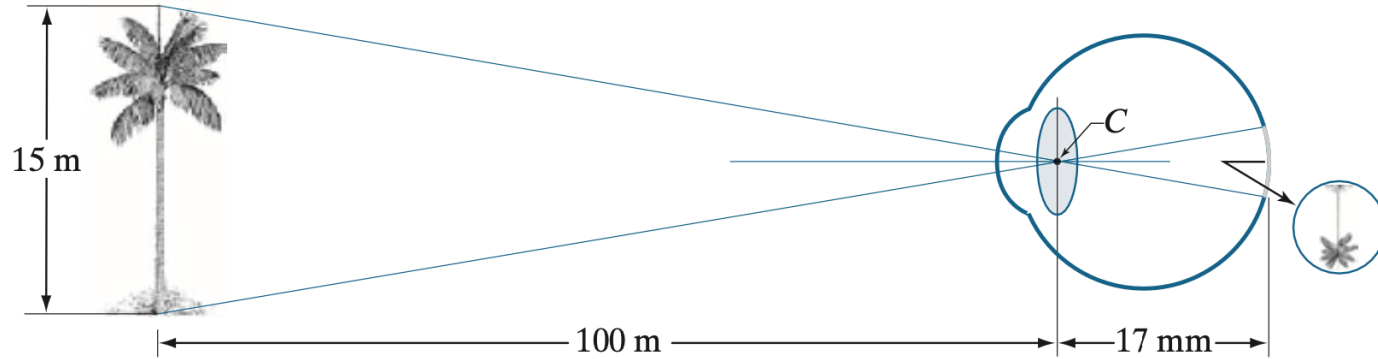
The slide features a light blue background. At the top, there are two horizontal bars: a thick teal one and a thin blue one. At the bottom, there are also two horizontal bars: a thin blue one and a thick teal one. In the lower-middle section, there are two short, thick, olive-green horizontal bars, one on the left and one on the right, positioned symmetrically.

Structure Of Human Eye



- The lens focuses light from the object onto the retina.
- The retina is covered with light-sensitive cells, including cone cells and rod cells.
- Cone cells are concentrated around the fovea and are highly sensitive to color.
- Rod cells are more widely spread and are highly sensitive to low light levels.

Image Formation In The Eye



- The muscles inside the eye can change the size of the lens, allowing the eye to focus on objects near or far.
- An image is focused onto the retina, which contains cone and rod cells, sending visual signals to the brain.

Image Formation In The Eye

The electromagnetic spectrum can be expressed in terms of wavelength, frequency, or energy. Wavelength (λ) and frequency (ν) are related by the expression :

$$\lambda = \frac{c}{\nu}$$

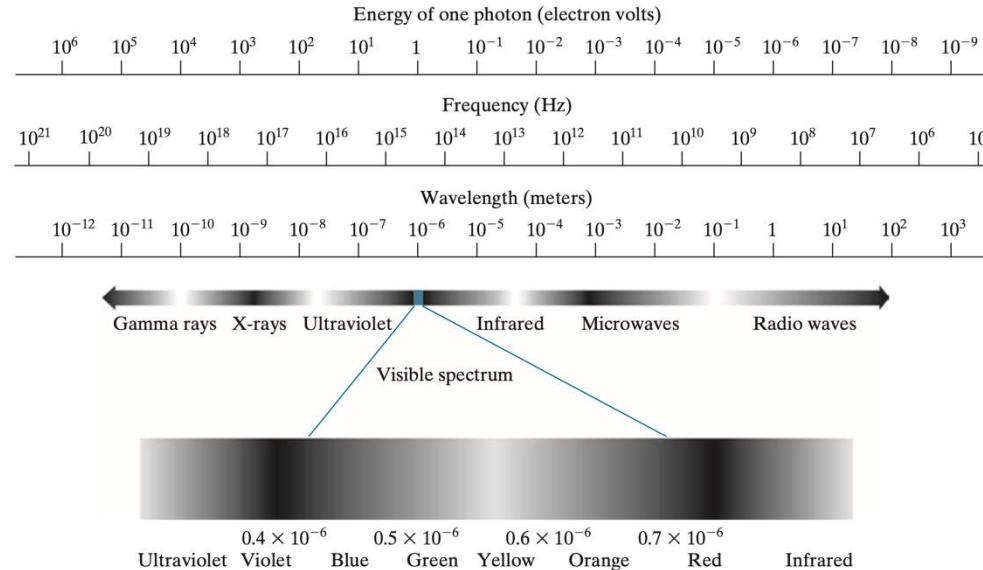
where c is the speed of light ($2.988 \times 10^8 m/s$).

The energy of the various components of the electromagnetic spectrum is given by the expression:

$$E = h\nu$$

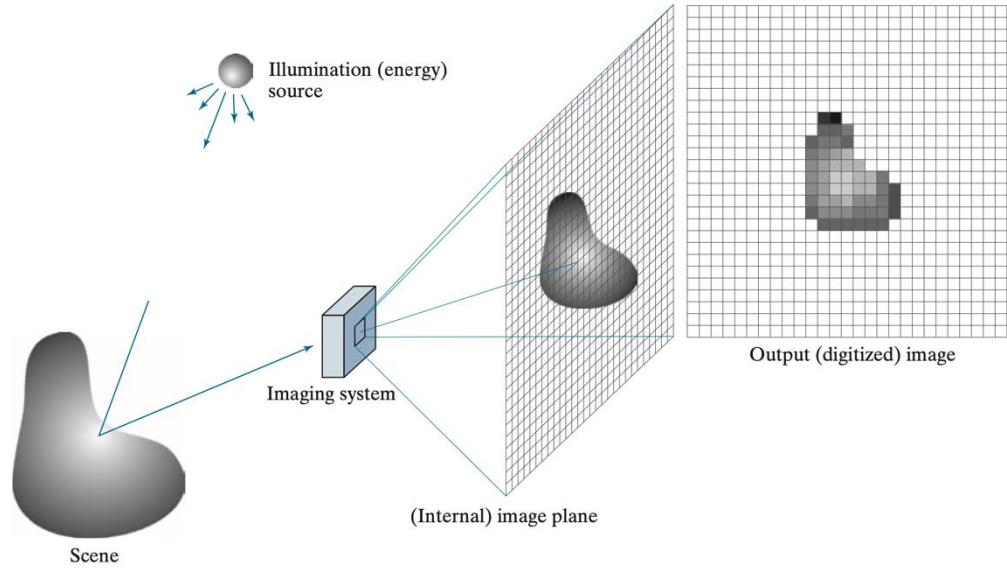
where h is Planck's constant

Image Formation In The Eye



Light is a type of electromagnetic radiation that can be sensed by the eye. For convenience, the color spectrum is divided into six broad regions: violet, blue, green, yellow, orange, and red

A Simple Image Formation Model



A Simple Image Formation Model

We denote images by two-dimensional functions of the form $f(x, y)$. The value of f at spatial coordinates (x, y) is a scalar quantity.

$$0 \leq f(x, y) < \infty$$

Function $f(x, y)$ is characterized by two components: (1) the amount of source illumination incident on the scene being viewed, and (2) the amount of illumination reflected by the objects in the scene. Appropriately, these are called the *illumination* and *reflectance* components, and are denoted by $i(x, y)$ and $r(x, y)$, respectively. The two functions combine as a product to form $f(x, y)$:

$$f(x, y) = i(x, y)r(x, y)$$

Where: $0 \leq i(x, y) < \infty$, $0 \leq r(x, y) \leq 1$

In the intensity (gray level) of a monochrome image, the common practice is to shift this interval numerically to the range $[0, 1]$.

A Simple Image Formation Model

$$0 \leq i(x, y) < \infty$$

$$0 \leq r(x, y) \leq 1$$

Why?

Representing Digital Images

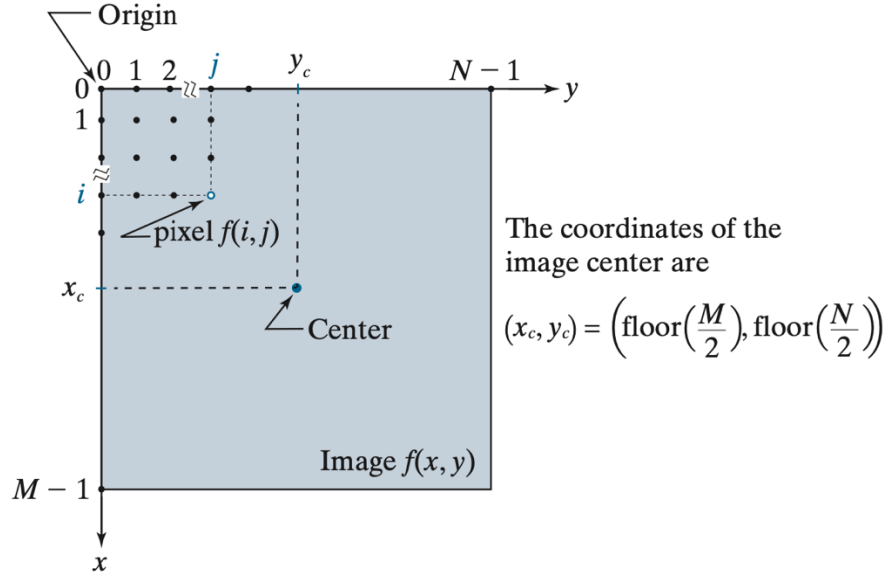
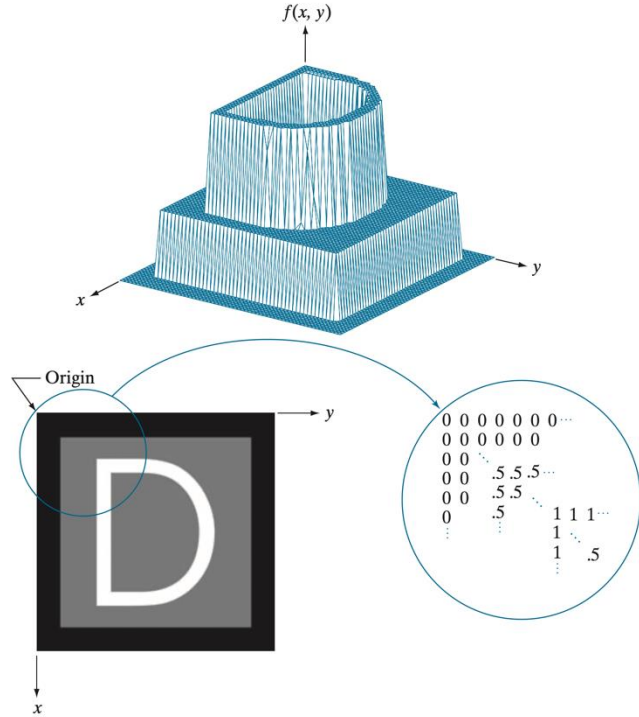
Let $f(x, y)$ represent a *continuous* image function of two continuous variables, x and y . Suppose that we sample the continuous image into a digital image, $f(x, y)$, containing M rows and N columns, where (x, y) are discrete coordinates. Where $x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, 2, \dots, N-1$

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

We can also represent a digital image in a traditional matrix form:

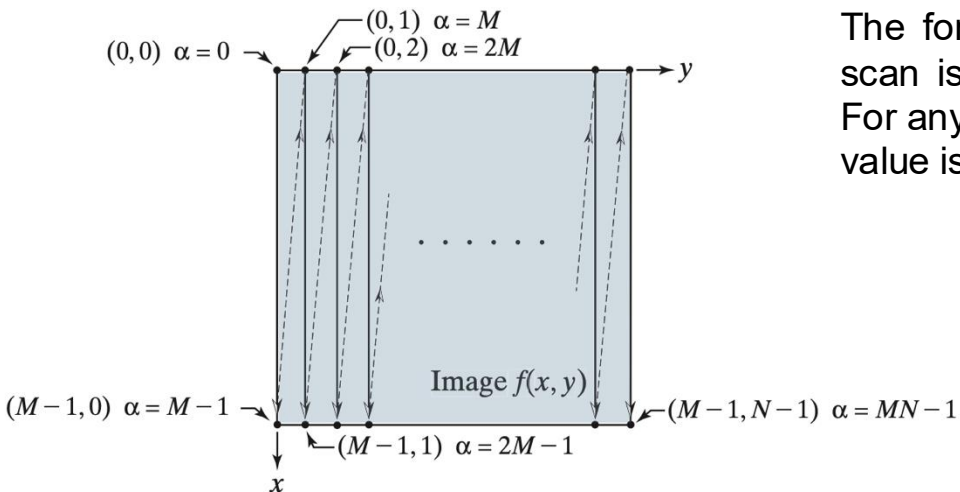
$$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \cdots & a_{M-1,N-1} \end{bmatrix}$$

Representing Digital Images



The *center* of an $M \times N$ digital image with origin at $(0, 0)$ and range to $(M - 1, N - 1)$ is obtained by dividing M and N by 2 and rounding *down* to the nearest integer. This operation sometimes is denoted using the floor operator

Representing Digital Images



The formula for generating linear indices based on a column scan is straightforward and can be determined by inspection. For any pair of coordinates (x, y) , the corresponding linear index value is

$$\alpha = My + x$$

$$x = \alpha \bmod M \quad y = (\alpha - x)/M$$

Dot Per Inch (DPI)

- Pixel count (e.g., 1024×1024) only indicates the number of pixels in the image matrix.
- Spatial resolution measures the smallest discernible detail, expressed as dots per inch (dpi) or line pairs per unit distance
- The same pixel count can correspond to very different spatial resolutions, depending on the physical size of the image.
- Example:
 - + 1024 pixels across 1 cm (0.39 in) $\rightarrow$ ~ 2600 dpi (very sharp).
 - + 1024 pixels across 1 m (39 in) $\rightarrow$ ~ 26 dpi (very coarse).

Suppose the image has: $M \times N$ pixels along the height (vertical direction). N pixels along the width (horizontal direction).
Physical size: H mm (height), W mm (width).

$$DPI_x = \frac{M}{W/25.4}, \quad DPI_y = \frac{N}{H/25.4}$$

Dot Per Inch (DPI)



(a) 930 dpi
(c) 150 dpi

(b) 300 dpi
(d) 72 dpi

Spatial And Intensity Resolution

- Refers to the smallest discernible change in brightness (gray level) in an image.
- Determined by the number of bits used for quantization.
- Common choices::
 - + 8 bits → 256 levels (standard in most digital images).
 - + 16 bits → 65,536 levels (used in medical and scientific imaging).
 - + 10 or 12 bits sometimes used in specialized devices.
- Higher intensity resolution allows smoother gradations of brightness and better detail in dark/bright regions.
- Practical visibility of intensity levels is influenced by noise, sensor limits, and human perception.

Spatial And Intensity Resolution

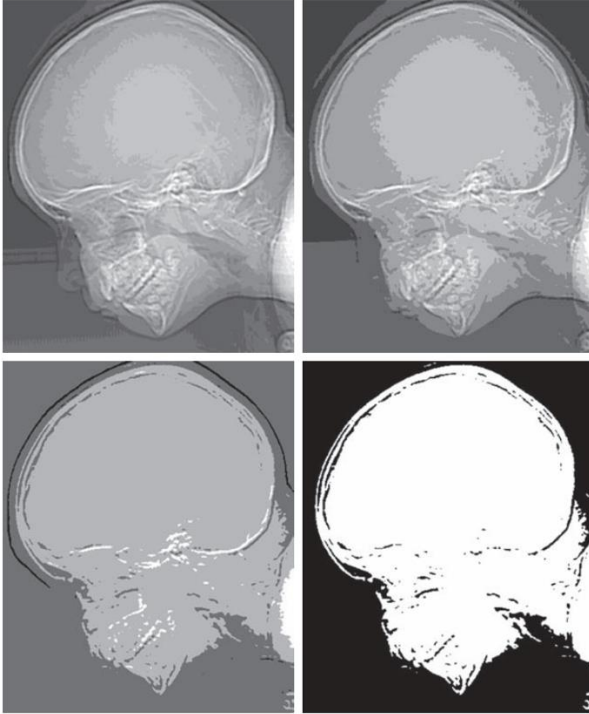


Image displayed in 16,
8, 4, and 2 intensity
levels.

Spatial And Intensity Resolution

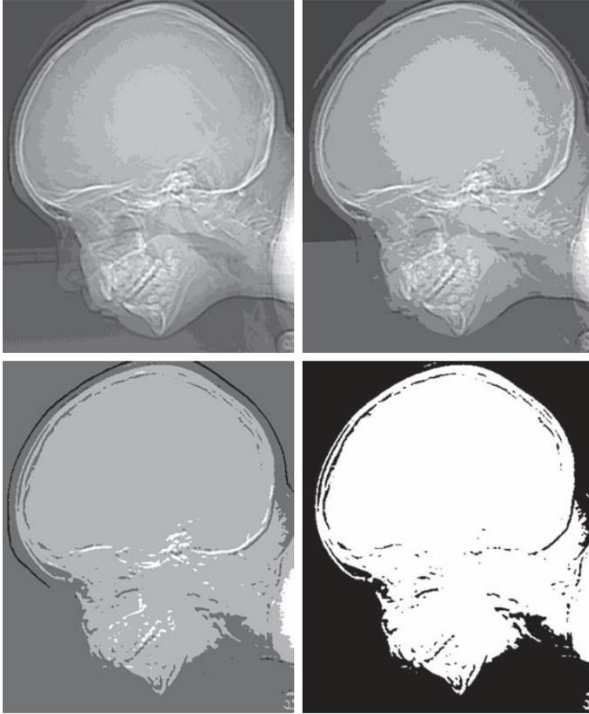


Image displayed in 16,
8, 4, and 2 intensity
levels.

Some Basic Relationships Between Pixels

In this section, we will explore several significant relationships between pixels in a digital image. In the subsequent discussion about specific pixels, we will use lowercase letters like p and q .

Neighbors Of A Pixel

A pixel p at coordinates (x,y) has two horizontal and two vertical neighbors with coordinates

$$(x + 1, y), (x - 1, y), (x, y + 1), (x, y - 1)$$

This set of pixels, called the 4-neighbors of p , is denoted $N_4(p)$

The four diagonal neighbors of p have coordinates

$$(x + 1, y + 1), (x + 1, y - 1), (x - 1, y + 1), (x - 1, y - 1)$$

and are denoted $N_D(p)$

These neighbors, together with the 4-neighbors, are called the 8-neighbors of p , denoted by $N_8(p)$

The set of image locations of the neighbors of a point p is called the *neighborhood* of p . The neighborhood is said to be *closed* if it contains p . Otherwise, the neighborhood is said to be *open*.

Neighbors Of A Pixel

	X, Y-1	
X-1, Y	X, Y	X+1, Y
	X, Y+1	

$N_4(p)$

X-1, Y-1		X+1, Y-1
	X, Y	
X-1, Y+1		X+1, Y+1

$N_D(p)$

X-1, Y-1	X, Y-1	X+1, Y-1
X-1, Y	X, Y	X+1, Y
X-1, Y+1	X, Y+1	X+1, Y+1

$N_8(p)$

Adjacency

Let V be the set of intensity values used to define adjacency. In a binary image, $V = \{1\}$ if we are referring to adjacency of pixels with value 1. In a grayscale image, the idea is the same, but set V typically contains more elements. We consider three types of adjacency:

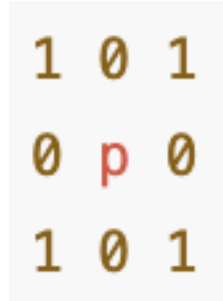
1. *4-adjacency*. Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.
2. *8-adjacency*. Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.
3. *m-adjacency* (also called *mixed adjacency*). Two pixels p and q with values from V are m -adjacent if:
 - (a) q is in $N_4(p)$ or
 - (b) q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V .

Adjacency

- 4-adjacency: pixels are neighbors if they share an edge.
- 8-adjacency: pixels are neighbors if they share an edge or a corner.
- m-adjacency (mixed): combines 4- and diagonal adjacency, avoids ambiguity in 8-adjacency.



4-adjacency



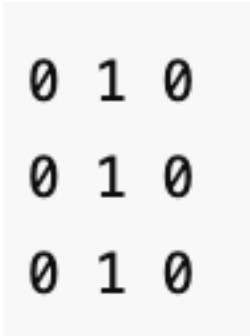
8-adjacency
m-adjacency



8-adjacency
Not m-adjacency

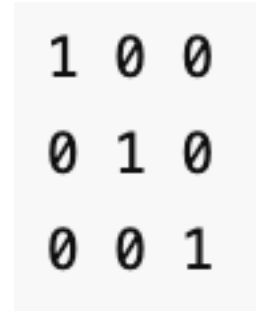
Connectivity

- Two pixels are connected if there exists a path of adjacent pixels between them.
- Connected component: maximal set of pixels where each pair is connected.
- Connectivity depends on chosen adjacency (4-, 8-, or m-).



0	1	0
0	1	0
0	1	0

4-connectivity



1	0	0
0	1	0
0	0	1

8-connectivity

Regions, Boundaries

- A region is a connected set of pixels with similar properties (e.g., intensity values in set V).
- Two regions are adjacent if at least one pixel from each region is adjacent.
- If not, the regions are disjoint.

0	0	0	0	0
0	1	1	0	0
0	1	1	0	0
0	0	0	1	0
0	0	0	1	0

Regions, Boundaries

- Boundary of a region = pixels in the region with at least one neighbor outside the region.
- Outer boundary: closed path surrounding the region.
- Outer boundary: closed path surrounding the region.

0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Outer boundary

0	0	0	0	0
0	1	1	1	0
0	1	0	1	0
0	1	1	1	0
0	0	0	0	0

Inner Boundary

Exercises

Exercise 1 — Neighborhood of a pixel Image (5×5 , 1 = foreground, 0 = background):

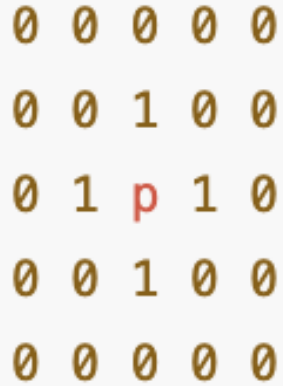


0	0	0	0	0
0	0	1	0	0
0	1	p	1	0
0	0	1	0	0
0	0	0	0	0

- a) List the 4-neighbors of p.
- b) List the 8-neighbors of p.
- c) Open vs. closed neighborhood (4-neighbors).

Exercises

Exercise 1 — Neighborhood of a pixel Image (5×5, 1 = foreground, 0 = background):



0	0	0	0	0
0	0	1	0	0
0	1	p	1	0
0	0	1	0	0
0	0	0	0	0

a) List the 4-neighbors of p.

Up (0,1 at row2,col3), Down (row4,col3=1), Left (row3,col2=1), Right (row3,col4=1).

So $N_4(p) = \{(2, 3), (3, 2), (3, 4), (4, 3)\}$ and all four have value 1.

b) List the 8-neighbors of p.

The four above plus diagonals: (2,2)=0, (2,4)=0, (4,2)=0, (4,4)=0.

So $N_8(p) = N_4(p) \cup \{(2, 2), (2, 4), (4, 2), (4, 4)\}$.

c) Open vs. closed neighborhood (4-neighbors).

Open 4-neighborhood: $N_4(p) \setminus \{p\}$ (the four around p, not including p).

Closed 4-neighborhood: $N_4(p) \cup \{p\}$ (includes p itself).

Exercises

Exercise 2 — Let's analyze adjacency between selected pixels (assume $V = \{1\}$ $V=\{1\}$).

1	0	1	0	0
0	1	0	1	0
1	0	1	0	1
0	1	0	1	0
0	0	1	0	1

- a) Consider pixel A at (1,1) and pixel B at (2,2).
 - 1. Are A and B adjacent under 4-adjacency?
 - 2. Under 8-adjacency?
 - 3. Under m-adjacency?

- b) Consider pixel C at (1,3) and pixel D at (3,1).
 - 1. Are C and D adjacent under 8-adjacency?
 - 2. Are they adjacent under m-adjacency?

- c) Pixel E at (3,3) and pixel F at (5,5).
 - 1. Are E and F directly adjacent under 8-adjacency?
 - 2. If not, is there a path between them using 8-adjacency?

Exercises

Consider the two image subsets, S_1 and S_2 in the following figure. Assuming that $V = \{1\}$, determine whether these two subsets are:

		S_1					S_2			
(a) 4-adjacent.	0	0	0	0	0	0	0	1	1	0
	1	0	0	1	0	0	1	0	0	1
(b) 8-adjacent.	1	0	0	1	0	1	1	0	0	0
	0	0	1	1	1	0	0	0	0	0
(c) m-adjacent.	0	0	1	1	1	0	0	1	1	1

Exercises

Assuming that $V = \{I\}$

- How many regions of 1's under 4-connectivity?
- How many regions under 8-connectivity?
- List the outer boundary pixels using 4-adjacency.
- Compare boundaries in 4-adjacency vs 8-adjacency.

0	1	1	0	0
1	1	0	1	0
0	1	0	1	0
0	1	1	1	0
0	0	0	0	0

Distance Measures

For pixels p , q , and s , with coordinates (x, y) , (u, v) , and (w, z) , is a *distance function* or *metric* if

(a) $D(p, q) \geq 0$ ($D(p, q) = 0$ iff $p = q$),

(b) $D(p, q) = D(q, p)$, and

(c) $D(p, s) \leq D(p, q) + D(q, s)$.

The D_4 distance, (called the *city-block distance*) between p and q is defined as

$$D_4(p, q) = |x - u| + |y - v|$$

The D_8 distance (called the *chessboard distance*) between p and q is defined as

$$D_8(p, q) = \max(|x - u|, |y - v|)$$

Distance Measures

The pixels with D_4 distance $\delta 2$ from (x, y) (the center point) form the following contours of constant distance:

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

The pixels with D_8 distance $\delta 2$ form the following contours of constant distance:

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Elementwise Versus Matrix Operations

For example, consider the following 2 X 2 images (matrices):

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

The elementwise product (often denoted using the symbol $\odot$ or $\otimes$) of these two images is

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

On the other hand, the *matrix product* of the images is formed using the rules of matrix multiplication:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Linear Versus Nonlinear Operations

One of the most important classifications of an image processing method is whether it is linear or nonlinear. Consider a general operator, $\mathcal{H}$, that produces an output image, $g(x, y)$, from a given input image, $f(x, y)$:

$$\mathcal{H}[f(x, y)] = g(x, y)$$

Given two arbitrary constants, a and b , and two arbitrary images $f_1(x, y)$ and $f_2(x, y)$, is $\mathcal{H}$ said to be a *linear operator* if

$$\begin{aligned}\mathcal{H}[af_1(x, y) + bf_2(x, y)] &= a\mathcal{H}[f_1(x, y)] + b\mathcal{H}[f_2(x, y)] \\ &= ag_1(x, y) + bg_2(x, y)\end{aligned}$$

Linear Versus Nonlinear Operations

Is the Max operator linear operator?

Linear Versus Nonlinear Operations

$$f_1 = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad f_2 = \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix}$$

and suppose that we let $a = 1$ and $b = -1$. To test for linearity, we again start with the left side of Eq. (2-23):

$$\max \left\{ (1) \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + (-1) \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right\} = \max \left\{ \begin{bmatrix} -6 & -3 \\ -2 & -4 \end{bmatrix} \right\} \\ = -2$$

Working next with the right side, we obtain

$$(1) \max \left\{ \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} \right\} + (-1) \max \left\{ \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right\} = 3 + (-1)7 = -4$$

Arithmetic Operations

Arithmetic operations between two images $f(x, y)$ and $g(x, y)$ are denoted as

$$s(x, y) = f(x, y) + g(x, y)$$

$$d(x, y) = f(x, y) - g(x, y)$$

$$p(x, y) = f(x, y) \times g(x, y)$$

$$v(x, y) = f(x, y) \div g(x, y)$$

Addition

Suppose that $g(x, y)$ is a corrupted image formed by the addition of noise, $h(x, y)$, to a *noiseless* image $f(x, y)$; that is

$$g(x, y) = f(x, y) + \eta(x, y)$$

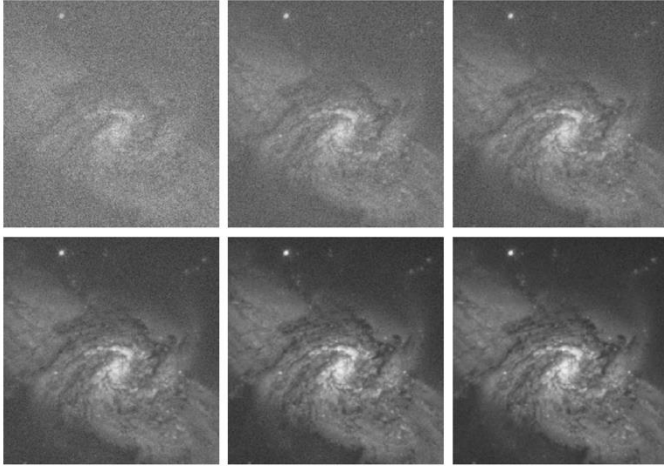
Where the assumption is that at every pair of coordinates (x, y) the noise is uncorrelated[†] and has zero average value. The objective of the following procedure is to reduce the noise content of the output image by adding a set of noisy input images, $g_i(x, y)$.

$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$

$$E\{\bar{g}(x, y)\} = f(x, y)$$

$$\sigma_{\bar{g}(x, y)}^2 = \frac{1}{K} \sigma_{\eta(x, y)}^2$$

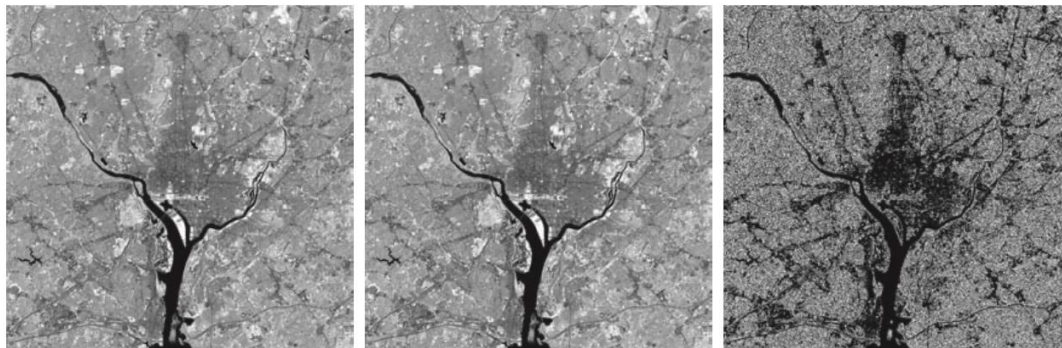
Addition



(a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)-(f) Result of averaging 5, 10, 20, 50, and 1,00 noisy images, respectively. All images are of size 566×598 pixels, and all were scaled so that their intensities would span the full $[0, 255]$ intensity scale. (Original image courtesy of NASA.)

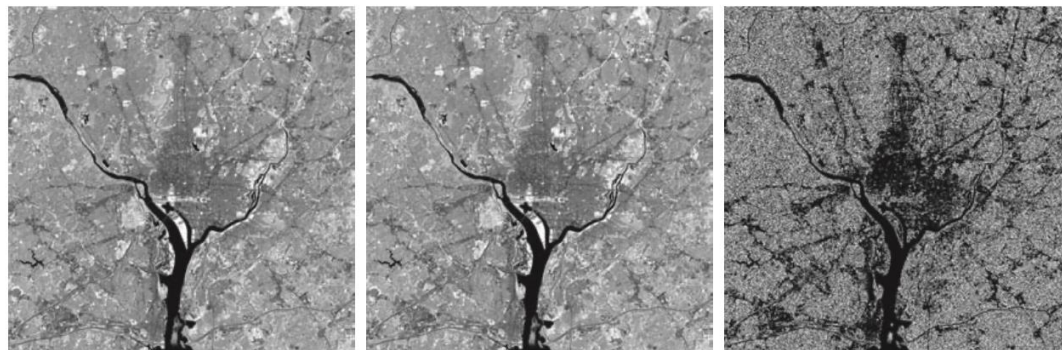
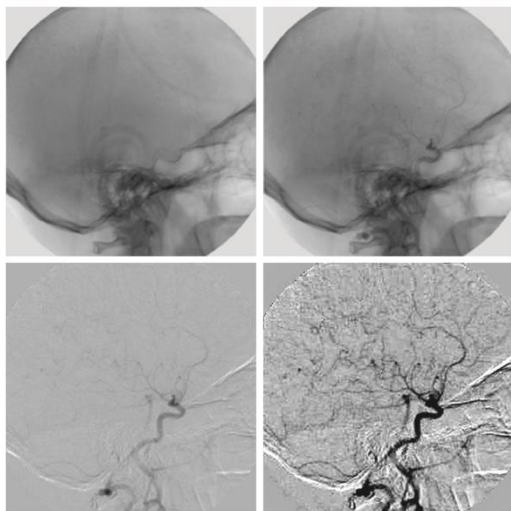
Subtraction

Consider image differences of the form: $g(x,y) = f(x,y) - h(x,y)$



Subtraction

Consider image differences of the form: $g(x,y) = f(x,y) - h(x,y)$



Multiplication And Division

- An important application of image multiplication (and division) is shading correction
- Shading correction is a method which can correct illumination inhomogeneities of the captured images.

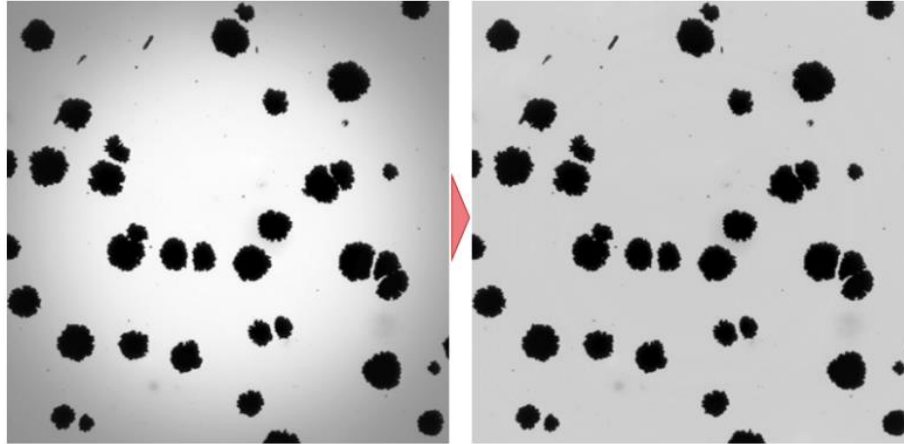


Image before and after applying the shading correction

Multiplication And Division

Suppose that an imaging sensor produces images that can be modeled as the product of a “perfect image,” denoted by $f(x,y)$, times a shading function, $h(x,y)$; that is,

$$g(x,y) = f(x,y) * h(x,y)$$

If $h(x,y)$ is known or can be estimated, we can obtain $f(x,y)$ (or an estimate of it) by multiplying the sensed image by the inverse of $h(x,y)$ (i.e., dividing g by h using elementwise division)

Multiplication And Division



(a) Shaded test pattern. (b) Estimated shading pattern. (c) Product of (a) by the reciprocal of (b)



(a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b)

Multiplication And Division

Given a digital image g resulting from one or more arithmetic (or other) operations, an approach guaranteeing that the full range of a values is “captured” into a fixed number of bits is as follows. First, we perform the operation

$$g_m = g - \min(g)$$

which creates an image whose minimum value is 0. Then, we perform the operation

$$g_s = K \lceil g_m / \max(g_m) \rceil$$

which creates a scaled image, g_s , whose values are in the range $[0, K]$. When working with 8-bit images, setting $K = 255$ gives us a scaled image whose intensities span the full 8-bit scale from 0 to 255

Basic Set Operations

A set is a collection of distinct objects. If a is an element of set A , then we write

$$a \in A$$

Similarly, if a is not an element of A we write

$$a \notin A$$

The set with no elements is called the *null* or *empty set*, and is denoted by $\emptyset$

A set is denoted by the contents of two braces: $\{ \circ \}$. For example, the expression

$$C = \{c \mid c = -d, d \in D\}$$

means that C is the set of elements, c , such that c is formed by multiplying each of the elements of set D by -1 .

Basic Set Operations

If every element of a set A is also an element of a set B , then A is said to be a *subset* of B , denoted as

$$A \subseteq B$$

The *union* of two sets A and B , denoted as

$$C = A \cup B$$

the intersection of two sets A and B , denoted by

$$D = A \cap B$$

Sets A and B are said to be *disjoint* or *mutually exclusive* if they have no elements in common, in which case,

$$A \cap B = \emptyset$$

The *sample space*, Ω , (also called the *set universe*) is the set of all possible set elements in a given application

Basic Set Operations

The complement of a set A is the set of elements that are not in A :

$$A^c = \{w \mid w \notin A\}$$

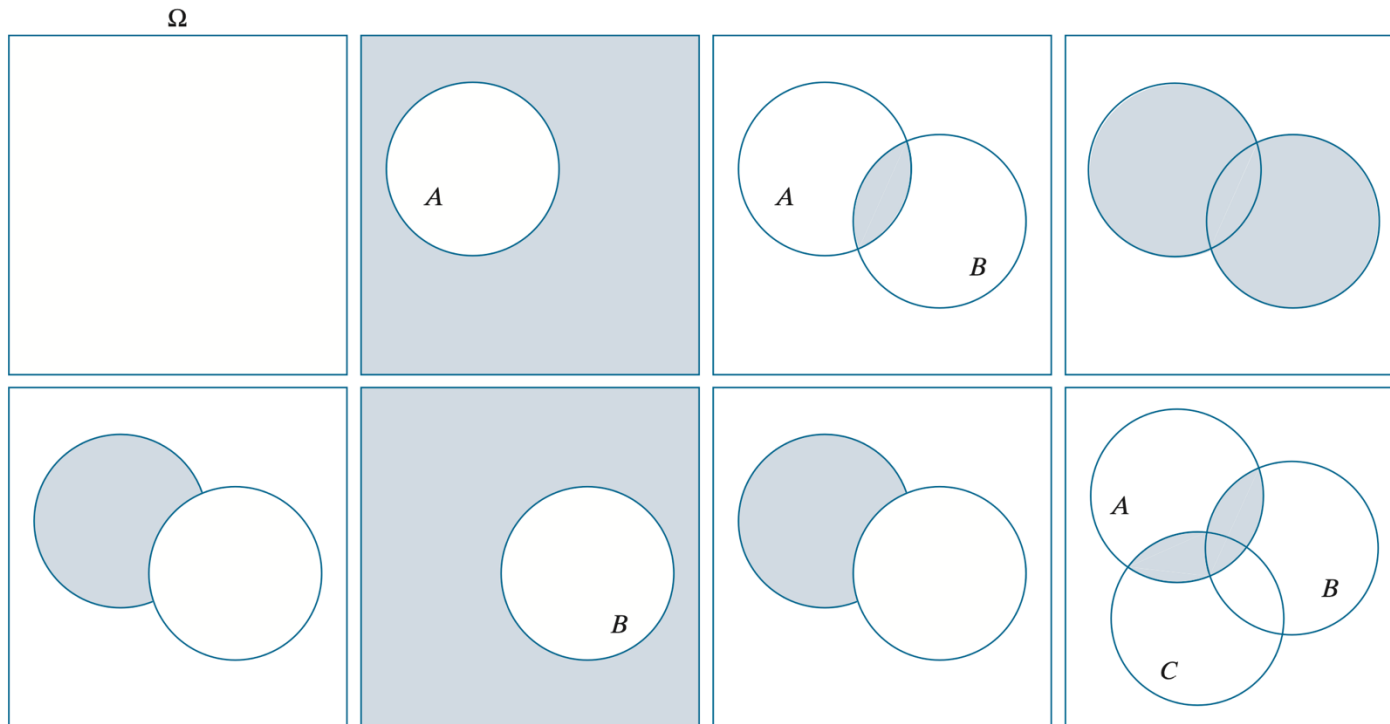
The difference of two sets A and B , denoted $A - B$, is defined as

$$A - B = \{w \mid w \in A, w \notin B\} = A \cap B^c$$

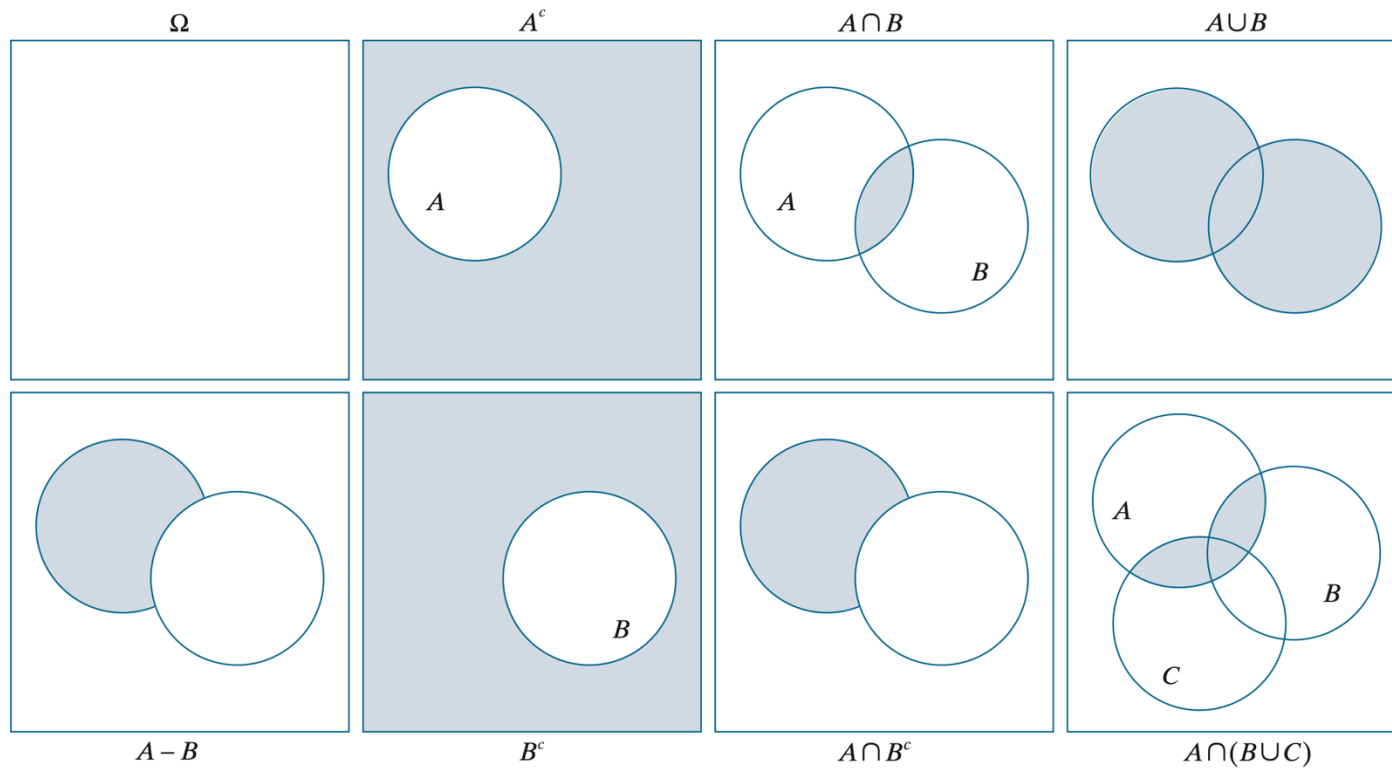
Basic Set Operations

Description	Expressions
Operations between the sample space and null sets	$\Omega^c = \emptyset; \emptyset^c = \Omega; \Omega \cup \emptyset = \Omega; \Omega \cap \emptyset = \emptyset$
Union and intersection with the null and sample space sets	$A \cup \emptyset = A; A \cap \emptyset = \emptyset; A \cup \Omega = \Omega; A \cap \Omega = A$
Union and intersection of a set with itself	$A \cup A = A; A \cap A = A$
Union and intersection of a set with its complement	$A \cup A^c = \Omega; A \cap A^c = \emptyset$
Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive laws	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
DeMorgan's laws	$(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$

Basic Set Operations



Basic Set Operations



Basic Set Operations

Let the elements of a grayscale image be represented by a set A whose elements are triplets of the form (x, y, z) , where x and y are spatial coordinates, and z denotes intensity values. We define the *complement* of A as the set

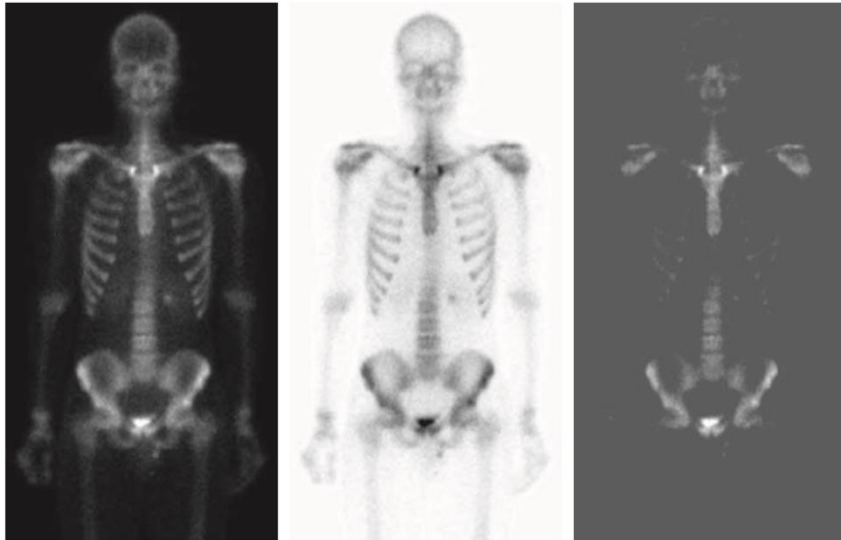
$$A^c = \{(x, y, 255 - z) \mid (x, y, z) \in A\}$$

Set operations involving grayscale images.

(a) Original image.

(b) Image negative obtained using grayscale set complementation.

(c) The union of image (a) and a constant image.



Basic Set Operations

The *Cartesian product* of two sets X and Y , denoted $X \times Y$, is the set of *all* possible ordered pairs whose first component is a member of X and whose second component is a member of Y . In other words

$$X \times Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}$$

Basic Set Operations

A *partial order* on a set S is a relation $\mathcal{R}$ on S such that $\mathcal{R}$ is:

- (a) *reflexive*: for any $a \in S$, $a\mathcal{R}a$;
- (b) *transitive*: for any $a, b, c \in S$, $a\mathcal{R}b$ and $b\mathcal{R}c$ implies that $a\mathcal{R}c$;
- (c) *antisymmetric*: for any $a, b \in S$, $a\mathcal{R}b$ and $b\mathcal{R}a$ implies that $a = b$.

Where, for example, $a\mathcal{R}b$ reads “ a is related to b .” This means that a and b are in set S , which itself is a subset of $S \times S$ according to the preceding definition of a relation. A set with a partial order is called a *partially ordered set*.

Let the symbol $\preceq$ denote an ordering relation. An expression of the form

$$a_1 \preceq a_2 \preceq a_3 \preceq \cdots \preceq a_n$$

Basic Set Operations

Of more interest to us later in the book are strict orderings. A *strict ordering* on a set S is a relation $\mathcal{R}$ on S , such that $\mathcal{R}$ is:

(a) *antireflexive*: for any $a \in S$, $\neg a \mathcal{R} a$;

(b) *transitive*: for any $a, b, c \in S$, $a \mathcal{R} b$ and $b \mathcal{R} c$ implies that $a \mathcal{R} c$.

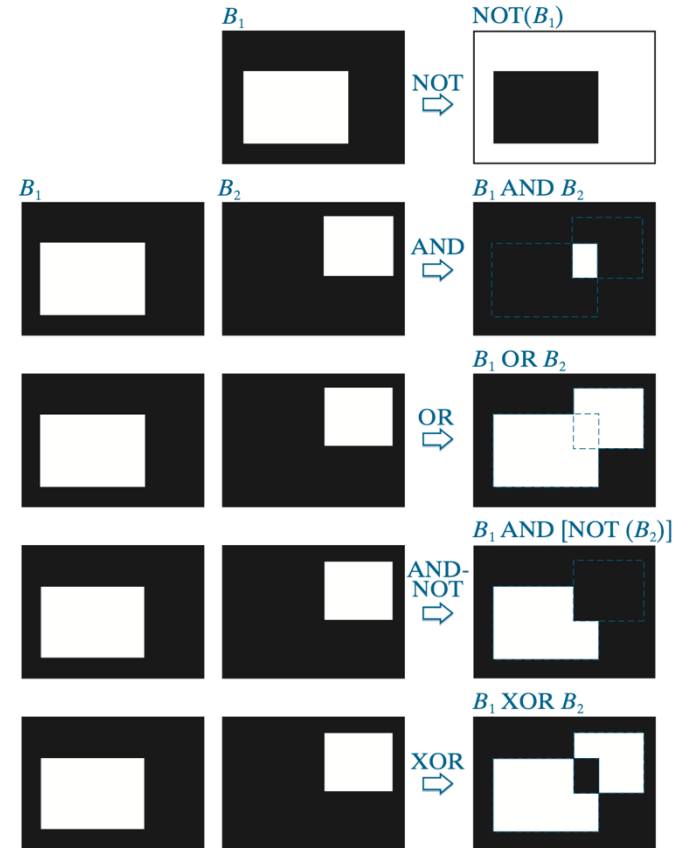
$$a_1 \prec a_2 \prec a_3 \prec \cdots \prec a_n$$

Logical Operations

Logical operations deal with TRUE (typically denoted by 1) and FALSE (typically denoted by 0) variables and expressions. For our purposes, this means binary images composed of *foreground* (1-valued) pixels, and a *background* composed of 0-valued pixels.

Logical Operations

a	b	$a \text{ AND } b$	$a \text{ OR } b$	$\text{NOT}(a)$
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

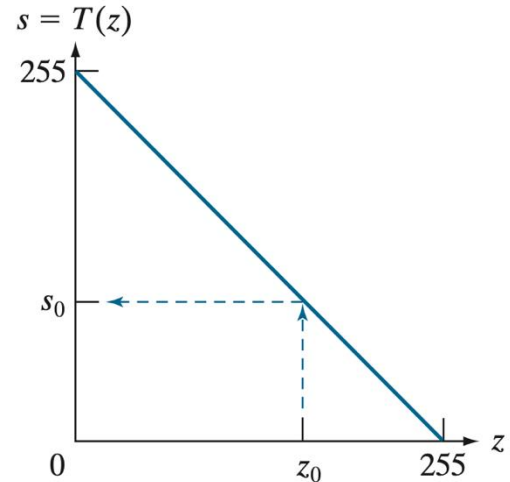


Single-Pixel Operations

The simplest operation we perform on a digital image is to alter the intensity of its pixels individually using a transformation function, T , of the form:

$$s = T(z)$$

where z is the intensity of a pixel in the original image and s is the (mapped) intensity of the corresponding pixel in the processed image.



Neighborhood Operations

Let S_{xy} denote the set of coordinates of a neighborhood centered on an arbitrary point (x, y) in an image, f . Neighborhood processing generates a corresponding pixel at the same coordinates in an output (processed) image, g , such that the value of that pixel is determined by a specified operation on the neighborhood of pixels in the input image with coordinates in the set S_{xy} .

$$g(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} f(r, c)$$

where r and c are the row and column coordinates of the pixels whose coordinates are in the set S_{xy} . Image g is created by varying the coordinates (x, y) so that the center of the neighborhood moves from pixel to pixel in image f , and then repeating the neighborhood operation at each new location

Neighborhood Operations

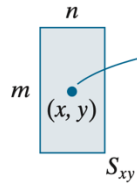
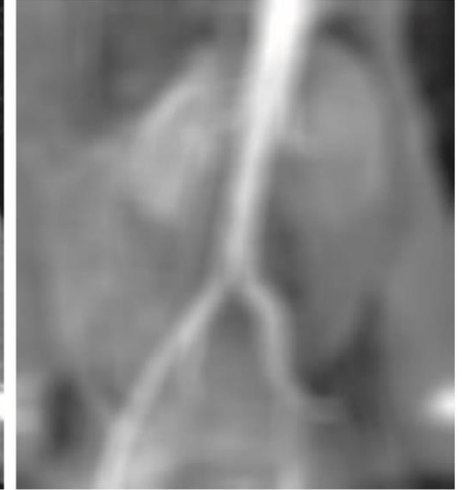
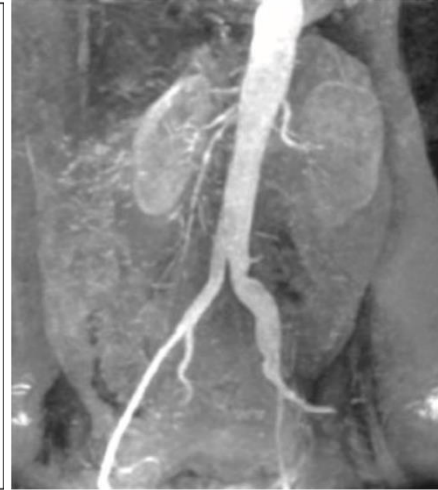


Image f

(x, y)

The value of this pixel
is the average value of the
pixels in S_{xy}

Image g



Geometric Transformations

We have

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

where (x,y) are pixel coordinates in the original image and (x',y') are the corresponding pixel coordinates of the transformed image.

Geometric Transformations

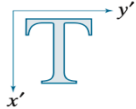
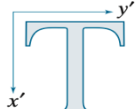
Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x \\ y' &= y \end{aligned}$	
Scaling/Reflection (For reflection, set one scaling factor to -1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= c_x x \\ y' &= c_y y \end{aligned}$	
Rotation (about the origin)	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x \cos \theta - y \sin \theta \\ y' &= x \sin \theta + y \cos \theta \end{aligned}$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x + t_x \\ y' &= y + t_y \end{aligned}$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x + s_v y \\ y' &= y \end{aligned}$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x' &= x \\ y' &= s_h x + y \end{aligned}$	

Image Intensities As Random Variables

We treat image intensities as random quantities in numerous places in the book. For example, let z_i , $i = 0, 1, 2, \dots, L - 1$, denote the values of all possible intensities in an $M \times N$ digital image. The probability, $p(z_k)$, of intensity level z_k occurring in the image is estimated as

$$p(z_k) = \frac{n_k}{MN}$$

Where n_k is the number of times that intensity z_k occurs in the image and MN is the total number of pixels. Clearly,

$$\sum_{k=0}^{L-1} p(z_k) = 1$$

Once we have $p(z_k)$, we can determine a number of important image characteristics. For example, the mean (average) intensity is given by

$$m = \sum_{k=0}^{L-1} z_k p(z_k)$$